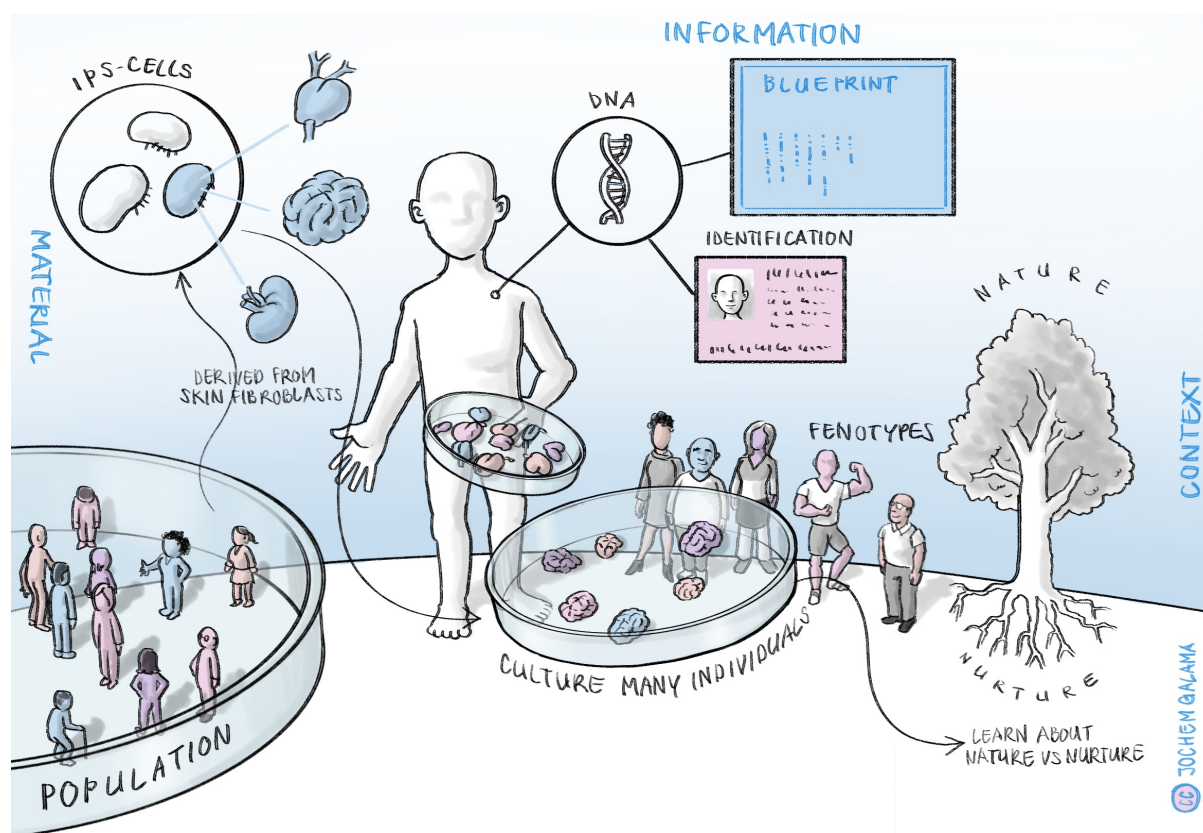


Convergence in a dish: Individualized CELLular models of disease diversity in the population



1. General information

1.a Applicants

<i>Name*</i>	<i>Institution</i>	<i>Department</i>	<i>Email</i>	<i>Tenured (i.e. fixed) position/ Tenure-track/ Other</i>	<i>For Erasmus MC applicants: clinician/ researcher / mix</i>	<i>Role in Flagship [Main Lead/Lead / Applicant]</i>
Joyce van Meurs	EMC	Internal Medicine and Orthopaedics & Sports Medicine Genomics Core Facility	j.vanmeurs@erasmusmc.nl	Tenured	Researcher	Main Lead and Lead EMC
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Gerjo van Osch	EMC/ TUD	[EMC]: Otorhinolaryngology and Orthopaedics & Sports Medicine [TUD]: Biomechanical Engineering	g.vanosch@erasmusmc.nl	Tenured	Researcher	Applicant

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Corresponding Lead: **Prof. Joyce van Meurs**

Planned start date of seed funding: *1-9-2022*

Primary Convergence theme: *Fundamentals of Health & Disease*

Secondary Convergence theme (*if applicable*): *Human Centered Technology and AI for Health*

Tertiary Convergence theme (*if applicable*): *Improving Health Journeys*

Quaternary Convergence theme (*if applicable*): Choose an item.

Keywords: *stem cells, population genetics, innovative cell models, precision medicine and health, advanced imaging, computational modeling*

1.b Summary (max 300 words)

Every person is unique and therefore we observe large differences between individuals in disease risk, treatment response and other human phenotypes. This diversity originates from an interplay between genetic and environmental variations. Optimal disease treatment or prevention for each individual requires a better understanding of the complex biology behind this interplay. Human disease modelling has been revolutionized by the arrival of induced pluripotent stem cells (iPSCs), which enables the generation of cellular and tissue models for each individual. Most iPSC studies have been conducted on a small number of donors (<10), but recent technical developments provide unique opportunities for large-scale applications (hundreds to thousands). Our Flagship **“Convergence in a dish” proposes to build a novel platform to develop, test and apply large-scale Individualized CELLular models (iCELL) across the normal population, to capture and understand (genetic) diversity between individuals in complex diseases, such as neurodegenerative disease, osteoarthritis and cancer.**

We will use “village cultures”, where iPSCs from many individuals are co-cultured using homogeneous conditions across the village. This cost-effective design will be coupled to innovative cellular/molecular phenotyping: high-throughput spatio-temporal imaging in combination with image-guided optogenetic selection of single cells harboring specific phenotypes followed by single cell RNA/DNA sequencing and proteomics. We will further improve the iCELL models by mimicking the natural environment of cells and tissues, as well as including intrinsic (e.g., genetic) and extrinsic (e.g., environmental) factors relevant for the diseases interest. Although iCELL can be used to study all diseases, we will focus on two examples: DNA-repair and fibrosis. We will create computational models based on game theory and machine learning to analyze individual disease characteristics and predict their response to treatment. In addition, applications of iCELL technology will be guided in an interactive co-creation process, where ethics, cost-benefit and public awareness will be evaluated together with valorisation opportunities.

1.c Public Summary (max 300 words)

Ieder individu is uniek en we zien daardoor grote verschillen tussen mensen in gevoeligheid voor ziekten en hun gevoeligheid voor behandeling. Voor veel ziekten begrijpen we de oorzaak van deze verschillen niet. Meer kennis is nodig voor een betere behandeling of preventie van ziekte toegesneden op het individu.

Om het effect van verschillen tussen mensen goed en efficiënt te kunnen onderzoeken zijn er nieuwe modellen nodig die de eigenschappen van een individu reflecteren in het laboratorium. Binnen ons flagship “Convergence in a dish”, gaan we geavanceerde technologieën combineren om celkweeksystemen te ontwikkelen voor ieder individu (iCELL) die breed toegepast kunnen worden voor verschillende ziekten. We testen het iCELL platform voor het onderzoeken van erfelijke aanleg en omgevingsrisicofactoren van ouderdomsziekten, zoals Alzheimer, osteoartrose en kanker. De

resultaten zullen vervolgens worden gebruikt voor het vinden van nieuwe behandelingsmethodes van ziektes, en het op maat maken van behandeling en preventie. iCELL zal worden ontwikkeld door een multidisciplinair team van wetenschappers van Erasmus MC, TU Delft en de Erasmus Universiteit. De toepasbaarheid van iCELL wordt gewaarborgd door een interactief ontwikkelproces, waarbij alle belanghebbenden betrokken worden en mogelijkheden voor valorisatie zullen worden geanalyseerd.

2. Relevance to Convergence Health & Technology mission (50%) – max. 2500 words for 2.a to 2.e

2.a Contribution to H&T mission and fit with H&T theme(s)

Our proposal encompasses all 3 convergence principles:

1. **Holistic life course approach:** We aim to build a platform to develop, test and apply individualised cellular models at large scale to capture and understand diversity between individuals. This platform can be used to study how individual health arises from a combination of genetic and environmental factors.
2. **Integrate health(care) innovation:** We aim to integrate novel technologies in iPSCs culture, high throughput imaging, advanced biomaterial engineering, genetics, single cell “omics”, game theory and machine learning to develop a system that can be used to increase understanding in pathology and pathogenesis of complex diseases and develop personalised treatments. Early assessment of possible output/spinoff is an integral part of the workplan.
3. **Responsible innovation:** Through co-creation, iCELL will evaluate and guide health innovation by exploring innovative applications. Our flagship will involve all needed expertise and stakeholders at predefined points during the development of the models. iCELL will be designed with ethical, legal and societal implications (ELSI) in mind to enable the responsible development, usability and continued sustainability of iCELL.

iCELL addresses specific challenges for the theme of fundamentals of health and disease:

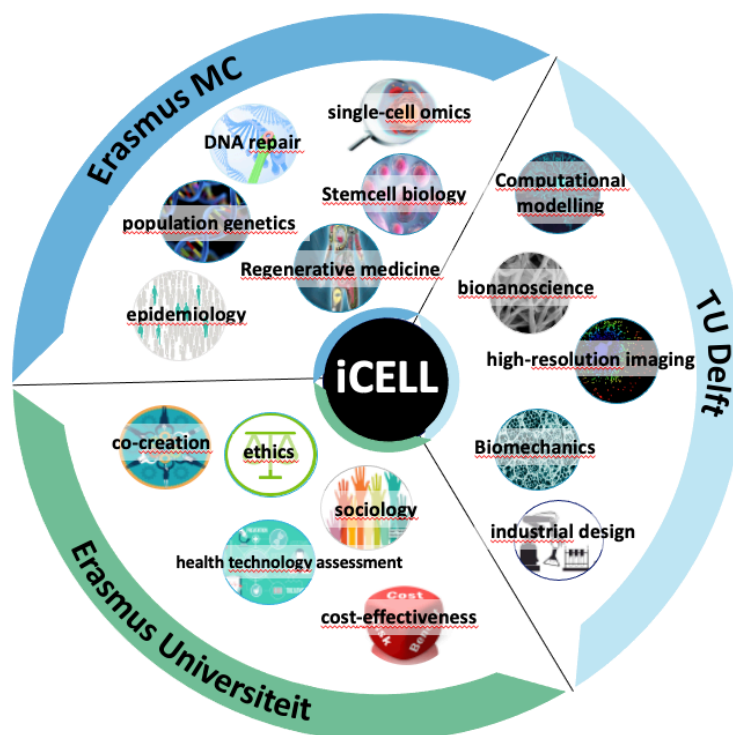
1. **Break down barriers between fields:** Our flagship requires close cooperation between biomedical and technical innovation and contains experts on stem cell technology, high resolution imaging, mechanobiology, population genomics and epidemiology, bioengineering, advanced statistical modelling and computational analysis. In order to streamline the strong potential towards health innovation, we will use participatory design to include the perspectives and expertise of all relevant researchers and stakeholders.
2. Our flagship will create a **platform to speed up translation** of relevant fundamental knowledge and technology into health innovation. iCELL models can be used to develop novel treatment and diagnostic strategies, and predict disease and treatment response.
3. Establishing iCELL models at population scale is important for many different diseases. We will focus on a few well-chosen **proof-of-concept examples**, where an enhanced insight into the drivers of variation between individuals has potential for broad impact:
 - The **biological pathway** DNA-damage repair and response (DDR): important in aging, neurodegeneration and cancer and exhibiting both tissue-specific and large inter-individual variability.
 - The **pathological process** of defective tissue repair/ fibrosis: a process of excessive connective (scar) tissue formation that occurs in many diseases, such as COPD and liver cirrhosis, is highly heterogeneous between individuals and can currently not be treated.

Our flagship perfectly fits in the *fundamentals of health and disease* theme, but applications and results from iCELL could help to improve *health journeys*, for example by understanding and predicting effects of treatments. In addition, with its strong link with revolutionary new technology and multimodal machine learning, game-theoretical modelling and data analysis, our flagship connects to the *Human centered technology & AI for health* theme.

2.b Synergy between disciplines

This flagship will leverage a unique combination of expertise within EMC, TUD and EUR as shown in the figure below.

As lead of the Rotterdam and Generation R studies, EMC and EUR have a rich history in population science and epidemiology, but functional readouts for these studies were limited to easily accessible tissues and cell types. Our consortium will develop the next level of population studies by combining complex population-based model systems of cell types of interest with powerful readouts developed at EMC and TUD, including single cell genomics, transcriptomics, proteomics, high resolution imaging and functional tracking.



The technology developed in this flagship can be used for any disease, but we will focus our proof-of-principle examples on **DNA repair and fibrosis**, representing key research spearheads at several departments of TUD and EMC. In addition, through co-funding two diseases will be examined: Alzheimers' disease and osteoarthritis. In addition, feasibility of high throughput applications of 'village in a dish' technology will be guided in an **interactive co-creation process**, where ethics, cost benefit and public awareness will be evaluated (expertise EUR) together, as well as possible valorisation opportunities. At present, worldwide application of this 'village in a dish' technology is in its infancy as successful development, implementation and application require input from a wide variety of disciplines, which has significantly slowed down progress in this research field. Here we have a unique opportunity, as all knowledge and expertise for development of the 'village in a dish' technology is present within the three Convergence institutes. Bringing these scientists together in this consortium will not only converge these expertise but will also create unique opportunities for

stem cell biologists, tissue engineers, geneticists, epidemiologists, (computational) modelers, data scientists and health technology assessment experts to address new scientific questions in the context of **functional population genomics**. With this unique constellation of scientists and expertise we think that this is the moment to invest and make this EMC, TUD and EUR flagship a world leading consortium on the 'village in a dish' technology.

2.c Long term sustainability plan

With this flagship we will build a sustainable platform for designing novel individualized cellular models at population scale that can be used for diverse purposes and possible user groups: fundamental and translational researchers, knowledge institutes (*e.g.*, TNO, part of this application) and the private sector. Fundamental and translational users share unique needs for the iCELL pipeline.

1. For fundamental scientific researchers: iCELL will be a powerful scientific platform that will facilitate and catalyze research aimed at understanding molecular pathways underlying diversity of biological processes in health and disease. Such deep understanding would facilitate the identification of novel genetic biomarkers and (personalized/stratified) treatment strategies that can be used to develop custom-made cellular models for scientific questions.
2. For translational scientists, deriving novel biomarkers or patient stratification models, that can be used (long-term) in predicting patients with poor treatment outcomes and adjusting treatment.
3. Private sector: during the first 5 years, knowledge on the large-scale culture of iPSC, as well single-cell phenotyping and innovative analyzing tools will be developed. Companies focused on drug development screening and diagnostic testing are expected to be interested in this technology and join our consortium as the Flagship progresses.

This flagship builds upon and brings together a number of innovative techniques for which proof-of-principle has been shown: i) Village cultures for iPSCs have been increasingly catching attention in the population genetics field, and recent reports show the first applications for neuronal and retinal cells¹⁻². ii) High spatiotemporal-resolution single cell identification platform coupled to single cell sequencing on multiple molecular levels developed by applicants Miao-Ping Chien, Daan Brinks and Jeroen Demmers³⁻⁴. iii) Mimicking the natural environment of the cell: Earlier work has evaluated 3 essential factors determining the microenvironment of the cell separately: micropatterning⁵, extracellular matrix composition⁶, and stiffness⁷⁻⁸. This flagship proposes to combine these 3 factors into a screening array to match the optimal natural environment to the studied cell. iv) Computational tools developed for the preprocessing and analysis of single-cell sequencing data⁹⁻¹², have achieved a high degree of maturity. We build upon those principles to develop custom machine learning models and include novel computational analysis principles. v) The Co-design concept has been widely used to design objects¹⁹ and telemedicine²⁰ but we intend to use it to accelerate applications in health-care from fundamental science, which is novel.

Since proof-of-principle has been shown, the feasibility of this program is high. The novelty lies within bringing these techniques together, applying them to different kind of data (for computational methods) and scaling them up to population-scale and translate them to clinical/societal application on the longer term.

Possible risk and contingency plan for long term sustainability

Risk	Contingency plan
No funding for further technological development – <i>low risk</i>	We are using core facilities (iPS, genomics, proteomics) in the EMC, as well as the FUNseq

	pipeline, that receive regular funding for services and technologies.
Slow translation to direct patient-care – <i>low risk</i>	Besides direct translation to patient-care, possible applications for the iCELL models are: Identifying new treatment targets, as well as cell models for screening for new drugs.
High throughput application in healthcare is dependent on automatisisation of iPSC-platform – <i>low risk</i>	Automation of iPSC generation is funded through a business plan and will take place independently from this Flagship application

Funding for long term sustainability:

The ECM and TUD core facilities (see sections 3.c) are centrally supported, ensuring long-term imbedding in the institutes.

We will pursue (and have already started to pursue) additional funding through additional grants (see also section 3b), examples are: Gravitational-NWO, KIC, NWA, TTW-OTP, TTW Perceptief and ZonMW Memorabel. Moreover, EU and US (NIH) funding options will be constantly monitored looking for calls aligned with the ideas and technologies of this flagship.

We expect that iCELL platform will attract many additional researchers and companies interested in using the developed technologies and researching the socio-economical aspects of its application. This has already happened during the writing of this flagship proposal, which resulted in the connection to Hieab Adams, who is interested in using the platform for studying neurodegenerative diseases.

2.d Visibility, attracting talents and new partners

The iCELL platform will be a major step for the Dutch research community to understand the origin of (genetic) diversity in health at a cellular level. This innovative technology will bridge the population genomic field with molecular and cell biology, innovative high-resolution single-cell monitoring tools, and computational models based on game theory and machine learning. iCELL will make us an attractive partner for future research projects and will build upon existing national initiatives to which several applicants of the flagship are connected (such as HEALTH-RI, X-Omics). The iCELL flagship is designed for growth by incorporating novel applications to enrich the cellular models and also to widen the number of applications. We will particularly reach out to diverse stakeholders to align their interest with the iCELL flagship and ensure an optimal further development and maximum use and benefit of the knowledge that will be gained. iCELL will invest in a comprehensive communication and dissemination effort that will be centered on: (a) building awareness of iCELL; (b) creating interest of (early career) researchers and other stakeholder; (c) public dissemination of the results generated within or with the use of iCELL; (d) promoting wider scientific and societal impact.

The following communication and dissemination activities will be part of iCELL, which are integrated in WP5 (Building an iCELL community):

1. A **website** that contains tailored information for the different possible stakeholder groups ranging from scientific information for researchers and professional healthcare providers to lay-man information for citizens, patients and companies.
2. Pro-active **dissemination program**:
 - News items and scientific output will be disseminated via website, linked-In, Twitter, Instagram, and via a newsletter within the iCELL community.

- Annual iCELL-convergence Workshop where we showcase our technologies and invite students, partners, international experts and possibly even the general public to participate
 - Showcase our work at top international conferences also via the organisation of specific symposia (e.g., at the TERMIS conference), and connect to partners in other ongoing projects and large-scale national/EU consortia we are involved in.
3. Set up a **training and education program**:
 - a. Include in our current teaching activities, the discussion of topics and technologies related to iCELL. For example, the BSc nanobiology program director is currently setting up a convergence minor, in which students from different programs work together on a health-related problem, each from their own specific expertise.
 - b. Set up an active program with bachelor and master projects co-supervised by at least two PI's from different institutes. We will recruit students at the annual nanobiology BEP/MEP event, a market where students can shop for an internship.
 - c. Propose to organize/expand specific programs for MSc students where at least the expertise of two institutions are represented (e.g., Biomedical Engineering, Nanobiology, ESHPM)
 4. Set up a scientific **co-creation community**, iCELL participants, students and interested 3rd parties, that jointly explore valorisation opportunities. Activities might include hackathons and grand challenges competitions.

2.e Long term societal impact

Contribution to healthcare, society and scientific impact:

The iCELL platform will provide unique opportunities to achieve a fundamental understanding of the initiation and progression of diseases from **pre-birth to end of life**, which is instrumental to the development of novel and improved disease diagnostics, prevention and treatment strategies. By establishing donor-specific disease models, the iCELL platform can be leveraged to identify the **right treatment for the right person, and ultimately come closer to the promise of precision medicine and personalised healthcare**. iCELL models are able to capture the sensitivity of an individual to a diverse range of diseases and are therefore essential to understand the reason for disease (intrinsic vs extrinsic factors) in each individual.

It is known that the prevalence of certain diseases, such as diabetes and stroke, is influenced by ethnicity. This can originate from both genetic differences and environmental factors. Despite this knowledge, most studies are still predominantly done in Caucasians. Village on a dish and iCELL models will offer the opportunity to take ethnic and socio-economic diversity across the population into account, with a huge potential towards **personalised medicine** and promoting much-needed **equality in healthcare**.

Expected outcome and valorisation opportunities:

The iCELL model will be developed in the **first 3 years** of the flagship with first proof of principle at the end of the **5 years** for – at least – the examples that we have chosen. Scientific impact in the first 3 years will be mainly at the scientific and technological level. We expect interest from companies for these models and will incorporate experts on valorisation opportunities in our co-creation monitoring team. Proof-of-principle studies on the effect of genetic variation and external factors on the cellular model will follow directly after. After this first phase of technological development, the applications of the individualised cellular models are numerous:

- Expanding the applications to other diseases; making the models more complex by (multicellular) differentiation.
- Societal impact of the technology lies in diverse applications such as drug target discovery, and screens for drug efficacy.

Technological Readiness Level:

Currently the village on a dish technology is on **TRL 2**. The concept of the technology has been described and experimental proof of concept (**TRL3**) is published in the field of neural differentiation. Regarding the single cell analyses these technologies are on **TRL3**. Experimental proof of concept is present for all technologies but we need to validate and optimise it for our application. Mimicking the natural extracellular environment is a strategy often used in the field of tissue engineering. The use of micropatterning and tissue-mimetic topographies (sometimes referred as “2.5D” cultures) to mimic natural extracellular micro-environments for iPSCs is less well developed and estimated to be at **TRL3** with experimental proof of concept that cells, when differentiated, will survive those extracellular micro-environments. We will need more basic knowledge on how biochemical and mechanical properties of the microenvironment together influence cell behaviour and how this can be used to make models that better mimic the *in vivo* situation. State of the art modelling techniques, such as game theory and machine learning, will be utilized for this purpose.

On all aspects of the project, we expect to reach **TRL5** by the end of the five years, where we will have proof of concept. By developing automated systems for cell culture, imaging and data analyses, we will be ready to implement the technologies for various applications in research laboratory.

3. Flagship proposal (25%) max 2000 words for 3.a to 3.d

3.a Ambitions

The overarching ambition of our Flagship “Convergence in a dish” is to **build a platform to develop, test and apply Individualized CELLular models (iCELL) at large-scale across the general population, to capture and understand diversity** between individuals. To reach this ambitious aim, we have built a consortium that uniquely combines expertise and innovative techniques from EMC, TUD and EUR using new forms of cooperation that cross boundaries between institutes and disciplines to create new perspectives and solutions. In this way, we will be able to address urgent and complex biological and societal challenges that stem from the current lack of individualized treatment for patients and accessibility for all.

3.b Aims and feasibility

Topic and aims

Genetic and environmental diversity between individuals influences risk to develop common diseases, differential treatment response and other human phenotypes, but is not well understood. Using human iPSCs (hiPSC) to study human complex traits may result in groundbreaking new knowledge through the intersection with population genomics. Recent work showed that hiPSCs can be used to investigate large-scale inter-individual variation for tissues that would be challenging (or impossible) to acquire *in vivo*¹³.

The aim of this flagship is to design individualized state-of-the-art cellular models **at large scale across the population** and build a series of activities to create an iCELL community. This approach will allow to capture and understand the **diversity between individuals** that reflects differences in sensitivity to diseases and response to treatment.

Concrete plan for attracting external funding

The applicants are actively involved in attracting external funding on topics related to this flagship. Between June 2021 and February 2022 the following project proposals have been submitted for evaluation, ~40% of them involving at least 2 different institutions: 4x NWO-OTP (Narcisi, Koenderink, Chien, de Mul; covering all WPs of this flagship), 3x NIH (Gonçalves, related to WP3), 2x NWO Vidi (Chien, Staňková; related to WP1 to WP3), 2x NWA-ORC (van Meurs, van Osch; related to WP1 to WP3), ERC synergy (Koenderink, related to WP2), HFSP young researcher grant and ENW-M (Brinks; related to WP2 & 3), 2x Horizon-EIC (Zadpoor, van Osch related to WP2), EU4H-2021-PJ2 (Staňková; related to WP2 and WP3), ROADMAP-NWO (van Meurs; related to WP3), 3x ZonMw-PSIDER (Gribnau, Narcisi; all related to WP1).

Possible risk and contingency plan for the project

Risk	Contingency plan
Limited access to iPSC lines (WP1) – <i>low risk</i>	We have several sources for iPSCs: i) iPS core facility, availability of >200 lines. ii) Large iPSC biobanks available (HipSci ¹⁴) iii) Ongoing generation of iPSCs for many diseases iv) The planned automation of iPSC pipeline will reduce the cost price and generate alternative opportunities, for example cells from the participants from the Rotterdam Study.
Complex to interpret and control cell response to multifactorial (biochemical and physical) inputs (WP2 & 3) – <i>low risk</i>	We will develop controlled substrates where different cues can be offered one-by-one and in combination. The DDR is also included as proof-of-concept, which is a well-studied and controlled process in cell culture.
Limited generalizability and interpretability of the analysis pipeline for modelling diversity and translation from cell models to patient (WP3 & WP4) – <i>intermediate risk</i>	We will limit this by regularization, cross-validation, and validation on independent co-cultured populations. Expertise with different computational approaches is present in the flagship.
Technical issues or insufficient expertise in the Flagship (WP1, 2 & 3) – <i>low risk</i>	The consortium integrates all necessary expertise close collaboration between the partners will resolve the issues raised by combining and scaling up the techniques that are already in place on smaller scale
Insufficient good-quality PhD students/postdocs available (all WPs) – <i>low risk</i>	PI's have a broad (inter) national network and are involved in teaching programs, which makes it possible to scout/recruit early-stage talented students.

3.c Work plan

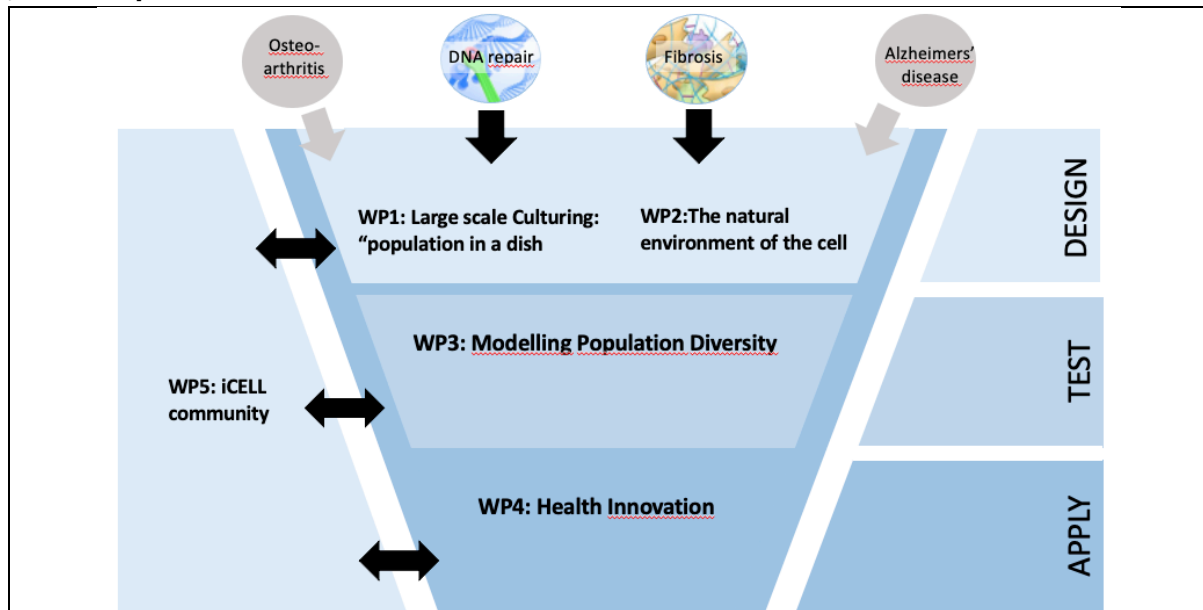


Figure 1: Workplan overview showing the WPs and two demonstrators (DNA repair and Fibrosis). In addition, two disease applications are part of the Flagship in the form of matching funding.

WP1: DEVELOP METHODOLOGY FOR CULTURING AND MONITORING A “VILLAGE IN A DISH”

AIM: Implement hiPSCs in genotype/phenotype studies of large cohorts iPSCs in a dish.

In this approach, a large number of hiPSCs are cultured and differentiated together in the same dish, and analyzed using sophisticated multimodal single-cell approaches.

TASKS:

(1.1): Implement village cultures for hiPSCs by:

- Identifying **criteria to select iPSC lines** for and apply the village culture on increasing number of individuals (up to 100).
- Implement single cell RNA-sequencing** (scRNA-seq) as a readout tool. Develop single cell read out for other molecular layers, such as proteomics (Proteomics Core-facility).

(1.2): Implement a large-scale high spatiotemporal-resolution **single cell imaging-sequencing platform** for hiPSCs to be able to select and/or measure (sub)cellular phenotypes of interest (Functional Single cell sequencing: FUNseq¹⁵). Select imaging conditions to use in WP3 that mimic the phenotype of interest (see demonstrators).

(1.3): Develop automated, modular systems for generation, expansion, and differentiation of hiPSCs to **scale-up the iCELL pipeline**. This will be done by a funded program. In the iPSC facility together with the department of industrial design.

The iPSC fulfilling the selection criteria (task 1.1) will be classified and chosen for further analysis (WP3). The optimized monitoring tools (task 1.2) will be used in WP2 and 3.

WP2: MIMICKING THE NATURAL ENVIRONMENT OF THE CELL

To enhance the translational potential of the iCELL platform to serve as an in vitro human disease model system, it is crucial to incorporate biomimetic micro-environmental characteristics that mimic the natural environment of the cell in the tissue.

AIM: Develop an array of “extracellular environments” that can be used to culture cells of a large number of individuals simultaneously

TASKS:

(2.1) Design and develop an **array of substrates** with different topographies, stiffnesses and biomolecular coatings ("extracellular environments") that can be used for single-cell phenotyping using high throughput microscopy.

(2.2) Develop the **FUNseq technology** (see task 1.2) in the different extracellular environments for the cellular phenotype of interest (see iCELL demonstrators).

(2.3) Select in vitro culture conditions to use in WP3 that **mimic the microenvironment** for the phenotype of interest (in this case fibrosis, see demonstrators) by comparing single-cell readouts (see task 2.2) with those in fibrotic tissue samples using neural network models.

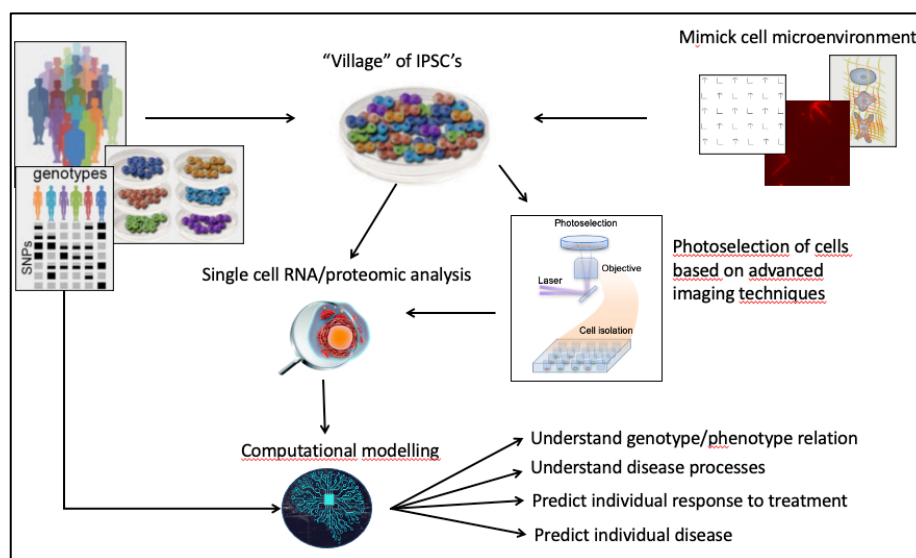


Figure 2: Technologies used to design and apply iCELL models.

WP3: MODELLING DIVERSITY IN CELLULAR PHENOTYPES TO CLARIFY THE LINK WITH DISEASE SUSCEPTIBILITY

AIM: To capture molecular signatures explaining diversity in clinically-relevant (cellular) phenotypes across the population.

TASKS:

(3.1): **Combine methodology of WP1 and WP2 and apply to a large number of iPSC donors (>100)** using scRNAseq or FUNseq. hiPSC lines carrying known high-impact mutations in the studied biological pathway will be included as positive control.

(3.2): **Develop interpretable machine learning models** to describe how donor molecular (genetic, transcriptomic) dynamics influence the chosen biological pathway (DDR capacity) and pathological process (fibrosis). We will use supervised learning, with regularization and game theoretic modelling will be used to further refine the models¹⁶. We will capture sub-population structure with dimensionality reduction¹⁷ followed by functional enrichment. Computational models will be validated against independently co-cultured iPSC populations, ensuring no overlap in donors.

(3.3) **Translate cellular to clinical phenotypes:** We will assess if the markers from 3.2 can also predict cases vs controls in large-scale studies (e.g., Alzheimers' disease or osteoarthritis, Rotterdam study), or if new models must be built to discern cases vs controls based on markers from 2.1.

(3.4) Apply gained knowledge to **clinically relevant questions** (identified in WP4): Computational models will be based on game theory¹⁶, machine learning¹¹ and phase-field approaches¹⁸.

WP4: HEALTH INNOVATION

AIM: Translation of the iCELL platform into research and healthcare applications. This will require **early participation from various stakeholders and early estimates of the potential value of its applications**. The application will be based on the EUnetHTA's principles of health technology assessment (EUnetHTA, 2016).

TASKS:

(4.1) Identify and assess **applications of the iCELL platform in health care**. Potential applications will first be supplied by iCELL partners and evaluated using the EUnetHTA core model for their potential value.

(4.2) **Co-design**¹⁹⁻²⁰ sessions with stakeholders are used to develop implementation pathways for future health-care applications as well as potential (ELSI) barriers, which will be studied and addressed during iCELL.

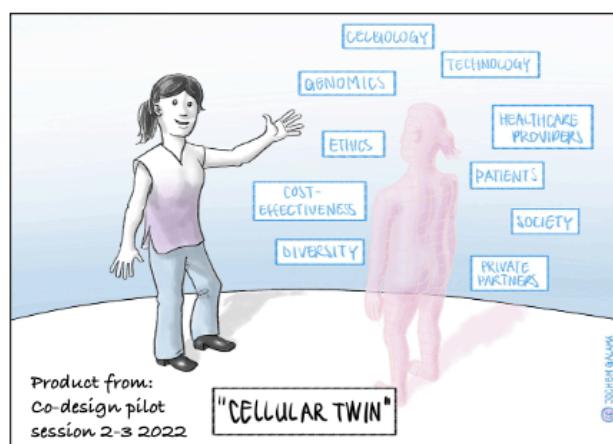
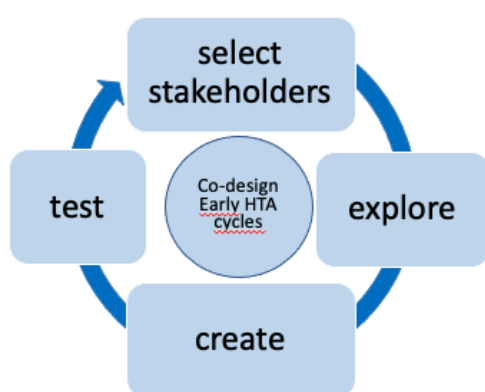


Figure 3: Co-design concept used in iCELL; the illustration is the result of a co-design pilot session where the "Cellular twin" emerged as the future dream-scenario.

(4.3) Regularly (every 6 months) estimate **stakeholder 'value'** to guide the implementation pathways using iCELL technologies developed and adjust WP activities to respond to stakeholder expectations during platform development.

WP5: Building an iCELL community

AIM and TASKS: (5.1) Organize flagship meetings and events; (5.2) Make a strategy for grant proposals. (5.3) Coordinate **MSc and PhD project plans** (5.4) **Train junior researchers** to build their own research line around iCELL and build career plan. Other concrete examples on how iCELL community building via communication and dissemination activities are presented in section 2.d.

iCELL DEMONSTRATORS:

In the first 3 years, we will focus on two concrete applications to establish and test the pipeline.

1. The **Biological pathway DNA damage repair (DDR)** is involved in many age-related diseases and cancer. Importantly, recent large-scale genetic studies have identified many (>150) common genetic variations in or near DDR-genes that influence human age-related phenomenon²¹. It is, however, unknown how the identified genetic variants influence the function of DDR. In this demonstrator, we aim to fill this knowledge gap. We will use a set of 100-200 cell lines, modify them to express fluorescent DDR proteins, challenge those cells with DNA damaging stimulus (X-ray irradiation) and apply the FUNseq pipeline to select DDR phenotypes²² followed by scRNA-sequencing and single cell proteomics. The responses to DNA damage will be analyzed using machine learning models and can be used to further refine DDR

network-modelling, for example by differentiating the IPS cells in the relevant cell types (e.g., fibroblasts or neurons).

2. The **pathological process of dysfunctional tissue repair, Fibrosis**, is part of the pathological process in many diseases and/or adverse treatment effects. We will evaluate how different combinatorial inputs of microenvironmental input parameters influence a fibrotic phenotype induced by known stimuli (TGF β , IL4/IL13). Multimodal imaging will be implemented to focus on differentiation state, cytoskeletal organization, mechano-signaling pathways, cell metabolism and physiology, and DNA repair.

We will study the effect of genetics and environment on fibrosis, develop a multi-modal pipeline that will be compared against baseline models built using molecular markers known to play a role in DNA repair and fibrosis.

At its completion, a variety of services will be provided to users of the iCELL platform:

1. Cost-effective high-throughput iPSC-culture expertise and methodologies using “village-in-a-dish”
2. High spatiotemporal-resolution single cell imaging platform applicable for high-throughput read-outs at population scale
3. Single-cell multi-omics technology at population-scale including transcriptomics and proteomics, but possibly also other omics-layers.
4. Innovative data analysis pipeline designed to apply novel predictive modelling.
5. Educational track (e.g., hands-on courses for bachelor, master and PhD students)
6. Specialised co-design team to assess translational and cost aspects at a very early phase.

Required facilities (already available)

Cell culture labs (TUD and EMC), iPSC core facility, genomics and proteomics core facility (all EMC), light microscope facilities (TUD), SEM, bio-AFM, 3D-printing and Kavli nanolab for nano/microfabrication, high-performance compute cluster of the Intelligent Systems department (all TUD). The EMC molecular genetics department houses many renowned experts in DNA repair.

3.d Alignment of the program with relevant national and international research agendas

The “village on a dish” flagship addresses at least **3 routes of the Dutch National Research Agenda** with the best overall fit to the route “Personalised Medicine: the individual at the centre”. Next to this primary route, the project also fits with route “Health care research, sickness prevention and treatment” and “Regenerative medicine: game changer moving to broad areas of application”. Our flagship also aligns to 3 of the focus areas of the **Europe EU program**: “health throughout the life course”, “tools, technologies and digital solutions for health and care including personalised medicine” and “health care systems”.

4. Consortium (25%) max 2000 words for 4.a to 4.d

4.a Track record of the consortium members

Name	Field of Expertise	Main topics and achievements in relation to the application	Publication number	BSc/MSc/PhD supervised	External funding (awarded or on going) in the last 5 years	WP involvement
van Meurs	Population genetics and integrative genomics	* Identification of new treatment for osteoarthritis and of novel mechanisms underlying genetic variants of complex diseases using integrative genomics. *Development of biomarkers and prediction tools for biological ageing *PI for genomics within the Rotterdam Study. *Head of population genomics unit of the genomics Core facility at EMC	>300	>50	NWO-VIDI, ZonMw, EU, NIH, ReumaNL	WP1,3,4,5
Gonzalves	Pattern recognition and bioinformatics	*Development of computational methods to learn from (multimodal) high-dimensional molecular data	>15	>20	NIH U54, Convergence Impulse, Varian-HollandPTC	WP1,3,4,5
Redekop	Health technology assessment and health services research	*Development and translation of innovative ideas about predictive biomarkers, imaging and artificial intelligence into viable products and services to improve healthcare efficiency. *Education of health technology assessment *Editor in chief of Health Policy and Technology	>200	>50	2x EU-Horizon, HartStichting	WP4,5
Narcisi	Stem cell biology and differentiation	* Identification of cellular pathways involved in stem cells differentiation	>35	>20	NWO-VENI, 2x ReumaNL, AO-foundation	WP1,4,5

		* Member TERMIS, ICRS, NBTE. * Secretary of the Dutch society for matrix biology (NVMB) * Associate editor for Frontiers.				
Koenderink	Cell and tissue biophysics	* Development of advanced imaging tools coupled with mechanical characterization from the molecular to material scale. * Collaboration with industry on biomaterials for food&health. * Member of several advisory boards (e.g., NWW-Science Domain) and editorial boards	>130	>90	NWO-VICI, Gravitation, ENW-XL, NWO-industrial Partnership program, Trombosestichting	WP2,3,4,5
van Osch	Connective tissue regeneration	* Identification of cellular processes during connective tissue degeneration and regeneration, in particular the role of stem cells, biomaterials and growth factors. * Public-Private partnerships * Past chair of European Chapter of Tissue Engineering and Regenerative Medicine International Society (TERMIS) * Active in several (inter)national societies, committees and editorial boards of scientific journals.	>250	>100	2x Horizon2020-MSCA-ITN, TKI-Health Holland, TTW-Perspectief	WP2,3,4,5
Ghazvini	Induced pluripotent stem cells	* Head/co-founder of Erasmus MC iPS core facility. * Member of The International Society for Stem Cell Research (ISSCR) and of the national facilities network of iPSC facilities	>50	>30	Michael J. FOX foundation, Dutch cancer society	WP1,5

		*PI of CorEuStem Network and International COREdinates Stem Cell Centres				
Chien	Multidisciplinary scientist in molecular biology, (bio)chemistry, advanced imaging, computational biology and single cell biology.	*Developer of single cell technology for cancer biology and diagnosis * PI at Oncode Institute and the head of UFO Biosciences *2x patents	>20	>40	KWF, VENI, Oncode Institute Junior Fellowship, CGC Junior Fellowship, Erasmus MC Fellowship, Ammodo Science Award	WP1,3,5
Gribnau	Stem cell biology	*Epigenetic mechanisms in gene expression regulation and translation to the clinic * Founder of the Erasmus MC iPS core facility *EMBO member, Oncode Institute member, Gravity NOCI Organs-on-a-chip consortium member * HFSP-career development award	>100	>150	NWO-VICI, ERC-consolidator	WP1,4,5
Brinks	Imaging physics	* Uses optical and molecular technology to investigate dynamic cellular physiology and computational technology to understand and control cellular processes * Founder of the Biomedical Intervention Optimization Delft AI lab (BIOlab)	>20	>30	NWO-startup ERC-starting	WP1,2,5
Pothof	DNA damage repair in aging	* Molecular and computational genetics of DNA damage and transcription stress in aging and pathology * Genomics, proteomics, transcriptomic * BSc nanobiology program director * Development of convergence education	>35	>30	NWO-open, ZonMw Memorable, NIH/NIA-P01, Horizon2020	WP3,4,5

Fratila-Apachitei	Cell-instructive surfaces	*Development of surfaces with multiple bio-functionalities *Coordinator of the cytocompatibility and bio-interfaces lab *Board member biomaterial-related journals (e.g., JFB and Biomedical Materials)	>75	>30	BMM, Interreg	WP2,4,5
Staňková	Game theory and its application in biology and healthcare	* Leading the Dynamic Game Theory team * Designing novel, evolutionary cancer therapies based on game theoretical models	50	>50	Horizon-ITN, NWO-Klein2, NWO Stairway to Impact award	WP3,5
Reinders	Computational biology, bioinformatics, machine learning	*Molecular classification and genetic network modelling * Single cell data analyses *Scientific Director of the TUD Bioengineering Institute, *Past scientific director of the Netherlands Bioinformatics Center	>300	>150	NWO Gravity, ICAI lab AI4.bio, 2x ZonMW Public-Private partnerships, EU-MARIE Curie Training Network	WP3,5
Adams	Medical doctor specializing in genetics	* Leads a multidisciplinary research laboratory aiming to improve our fundamental understanding and clinical care of complex disorders *Initiated multiple global collaborations *Editor for Alzheimer's and Dementia	>100	>30	NWO-open, EU-MARIE Curie	WP1,3,4,5
Demmers	Mass spectrometry based quantitative proteomics and biochemistry	*Director and founder of the proteomics core facility at EMC, which part of the NWO Roadmap for Large-scale Infrastructure Facilities *Theme leader in the NWO Proteins@work consortium and a board member of the National Proteomics Platform professional society *Demmers Lab is part of the NWO Roadmap for	>170	>60	NWO-proteins @work	WP1,5

		Large-scale infrastructure Facilities					
Zadpoor	Biomaterials & Tissue Biomechanics	* Founding director of the Additive Manufacturing Laboratory at TUD * Development of advanced manufacturing techniques for metamaterials with unprecedented mechanical, physical, or biological properties * World recognized expert in origami and kirigami-based (bio)materials * Early career award from the European Society for Biomaterials	>100	>100	ERC, NWO-VENI, NWO-VIDI,	WP2,4,5	
van Rooij	Population genomics analysis, bioinformatics, biostatistics	* Clinical or scientific applications for genomics data * Involved in >30 projects requiring complex/big data modelling	>50	>30	-	WP1,3,4,5	
Goossens	Physical Ergonomics and Industrial Design	* Chair of the Delft Health Initiative * Head of department of Industrial design	>40	>40	2x NWO	WP1	
Kenjeres	Multi-scale multi-physics transport phenomena in bio-medical applications	* ICHMT and EURO THERM Committee Member * Research Fellow of the Royal Netherlands Academy of Sciences and Arts	>150	>200	Dutch Heart Foundation, FF4EuroHPC/ DiTAIDD	WP3,4,5	
De Mul	Digital Innovation	* Adoption and embedding of new technologies in health care practice * Patient participation in health technology development * Co-creation development * Development and coordination of the	>40	>100	EU-Horizon	WP4,5	

		bachelor program in Health Sciences				
vandeKerckhove	Co-design methodology for digital health	*Development of stakeholder selection * Development of a scientifically robust method to effectively co-design digital health * Published a book about patient experience "The Self Hologram: Awakening a philosophy of life"	5	5	Educational Grant - EUR	WP4,5
Huang	Ethical and philosophical issues in emerging healthcare technologies	* Initiation of several interdisciplinary collaborations on the ethics of digital twins for personalised healthcare (EMC and TUD)	7	-	Further Funding Grant from the Society of applied philosophy, Open Mind call Convergence	WP4,5
Ikram	Epidemiology of neuro-degenerative disease	* Principle Investigator of the Rotterdam Study * Head of department of Epidemiology, EMC	n.a.	n.a.	n.a.	Advisory board
Powell	Complex genetics and cellular genomics	* Director of the Garvan-Weizmann Centre for Cellular Genomic * Genomic mechanisms (complex human diseases) * founding director of SeqBio	n.a.	n.a.	n.a.	Advisory board
Stoop	*In vitro diagnostic tests	*development of in vitro and in vivo model systems for the investigation of fibrosis and metabolic overload driven pathologies * Holds multiple patents	n.a.	n.a.	n.a.	Advisory board

4.b Expertise of the consortium members in relation to the aims and objectives of the program

This multidisciplinary flagship aims to tackle scientific, technological and societal problems from different perspectives.

The specific aim of the flagship is to develop a platform to design, test and apply iCELL models: individualized state-of-the-art in vitro models, that capture diversity between individuals in sensitivity to diseases and response to treatments. We can achieve this aim only with a broad consortium composed of members with complementary expertise. We therefore assembled a team of scientists, technology and health economy/policy experts that, together with the advisory board (Ikram and Powell), will be able to reach all the 4 main objectives described in the WPs.

The iPS core facility (EMC; Ghazvini) together with experienced scientists in stem cell biology (EMC; Narcisi, Gribnau) will be essential to develop methodologies to cost-effectively co-culture iPS cell lines. This will be combined with real-time single cell “omic” phenotyping of cells, which will be done by a team of scientists already collaborating on this topic and adding expertise on single cell biology & technology (EMC; Chien), high resolution and high-throughput imaging (TUD; Brinks and EMC; Chien), proteomics (EMC; Demmers; proteomics CoFa) and data science (TUD; Reinders, Gonçalves).

In order to mimic the natural environment of the cell, key regenerative medicine experts (EMC; van Osch) will work closely together with experts on bionanoscience (TUD; Koenderink) and material scientists (TUD; Zadpoor, Fratila-Apachitei) building on a recently started project in the Convergence IMPULSE program as well as Medical Delta Programme and extending this with longitudinal 3D imaging technology (TUD; Brinks).

The developed models will be used to study origins of heterogeneity across the population, including the effect of genetic variation (EMC; van Meurs, van Rooij, Adams) on the proof-of-principle examples that we have chosen: DNA-repair (EMC; Pothof, Adams) and tissue repair (EMC; van Osch, Narcisi and TUD; Koenderink). The dysregulations of those processes often lead to a various spectrum of diseases, such as cancer, fibrosis and neuronal loss. The resulting information (individual heterogeneity of response) will be used in game-theoretic and machine-learning models to predict organ/tissue disease and/or response to treatment by computational experts (TUD; Kenjeres, Staňková, Gonçalves).

The department of developmental biology (EMC; Gribnau), the iPS-facility (EMC; Ghazvini) and the Faculty of Industrial design (TUD; Goossens) will team up to develop automation of culture and differentiation of iPS cell lines, which is expected to be ready at year 3 of the proposal. To effectively implement improvements in health(care) based on the resulting innovative technology platform, we will address economic feasibility and societal acceptance as an integral part of our joint endeavour (EUR; Redekop, de Mul, Vandekerckhove, Huang).

4.c Involvement of partners

Section 4b describes all the different expertise of the consortium partners and how they will be used in this consortium. The applicants are connected to a wide external network of scientific collaborations for example through (inter)national consortia in the field of population genetics and genomics (*e.g.*, CHARGE, GO, many more), national infrastructure programs (*e.g.*, XOMICS, HEALTH-RI, BBMRI), and several Gravitation programs (*e.g.*, BaSyC).

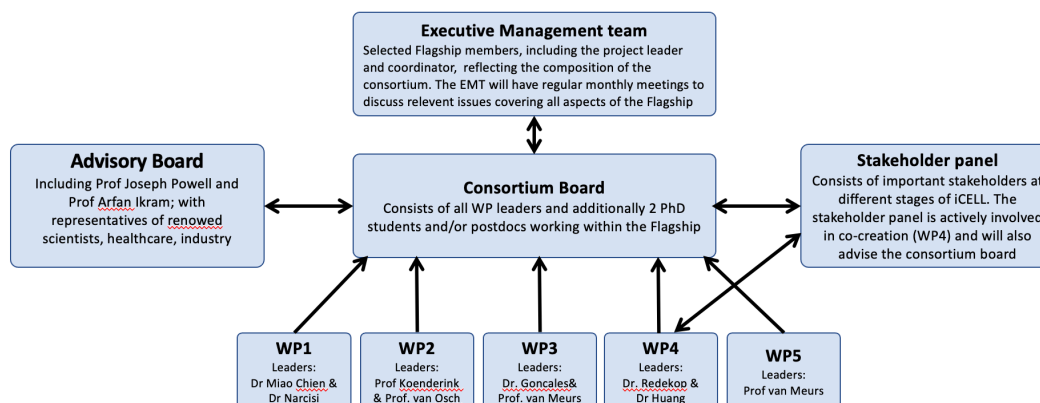
Several applicants have extensively collaborated with private partners (*e.g.*, Koenderink, van Osch, Reinders, Redekop) which ensures excellent opportunities for valorisation. This is also exemplified by the involvement and letters of support from Mimetas and TNO.

Important additional external partners for iCELL are also clinicians, with whom many of the applicants have close connections. These will be part of the stakeholders panel and where possible involved in follow-up projects.

4.d Governance, organization and collaboration strategy of the consortium

iCELL will be **coordinated** by Prof. dr. Joyce van Meurs (EMC), with support from the program coordinator dr. Narcisi Roberto (EMC). The **consortium board** will be the decision-making body of the consortium, including scientific and financial management. The consortium board will consist of the project lead, coordinator, and all WP leaders. We have chosen for a mix of senior and more junior WP leads to stimulate career development of talented junior researchers. Junior PI's are backed up by a senior member of the same group. The **consortium board** will be responsible for monitoring and securing the progress of the project, including organization of meetings, conference calls regularly to monitor and plan future strategy. This will include a kick-off meeting within 3 months of the start of the flagship and at least 2 overall flagship meetings per year.

Governance Structure iCELL



The **executive management team** consists of selected members to broadly reflect the composition of the consortium, who will meet extra 2 times a year to discuss progress, strategy and issues concerning the flagship program. An **Advisory Board** will be composed of national and/or international leading experts to provide advice to the EMT on scientific strategy and clinical, economical and societal valorization, and will be invited to attend yearly consortium meetings. The co-creation concept which is central in WP4 ensures the involvement of all necessary expertise in this project. We will compose a **stakeholder panel** which consists of important stakeholders at different stages of iCELL design, application and health innovation. This will include healthcare workers (physicians), patient/population representatives, ELSI experts (*e.g.*, HEALTH-RI), valorisation expert (*e.g.*, Health-Holland). We have already initiated a pilot co-creation session in the preparation towards this flagship and have the intention to use it throughout the project. In addition, WP5 is dedicated at building an iCELL community, the strategy and tasks have been described in detail in section 2d.

Annexes

Budget R&D overview: kostensoorten per type onderzoek (hele project periode)

R&D Overview: kostensonten per type onderzoek (hele project periode)																																																																																																																																																																																																																																																																																	
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	Partner die de kosten maakt	Functie	Aantal fee	Startdatum	Einddatum	Inschaling	Trade	Maandbedrag		Kosten soort (hele project periode)		2022		2023		2024		2025		2026		2027		2028		2029		2030		2031		2032		2033		2034		2035		2036		2037		2038		2039		2040		2041		2042		2043		2044		2045		2046		2047		2048		2049		2050		2051		2052		2053		2054		2055		2056		2057		2058		2059		2060		2061		2062		2063		2064		2065		2066		2067		2068		2069		2070		2071		2072		2073		2074		2075		2076		2077		2078		2079		2080		2081		2082		2083		2084		2085		2086		2087		2088		2089		2090		2091		2092		2093		2094		2095		2096		2097		2098		2099		2100		2101		2102		2103		2104		2105		2106		2107		2108		2109		2110		2111		2112		2113		2114		2115		2116		2117		2118		2119		2120		2121		2122		2123		2124		2125		2126		2127		2128		2129		2130		2131		2132		2133		2134		2135		2136		2137		2138		2139		2140		2141		2142		2143		2144		2145		2146		2147		2148		2149		2150		2151		2152	

Annex1: deel 2: Financiering-bijdragen per partner per jaar

FINANCIERING - Bijdragen per partner per jaar

In Euro		2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner
		In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	
Onderzoekorganisaties																
1	Erasmus MC		€ 68.884		€ 265.835		€ 244.029		€ 180.601		€ 66.208		€ 0		€ 825.557	€ 825.557
2	TU Delft		€ 73.595		€ 225.200		€ 166.907		€ 101.636		€ 38.946		€ 0		€ 606.284	€ 606.284
3	Erasmus University		€ 17.191		€ 52.603		€ 35.770		€ 0		€ 0		€ 0		€ 105.564	€ 105.564
4	Convergence bidr	€ 135.000		€ 400.000		€ 400.000		€ 400.000		€ 400.000		€ 265.000		€ 2.000.000		€ 2.000.000
...	Organisatie 4													€ 0	€ 0	€ 0
...	Organisatie 4													€ 0	€ 0	€ 0
...	Totale bijdrage ont	€ 135.000	€ 159.669	€ 400.000	€ 543.638	€ 400.000	€ 446.706	€ 400.000	€ 282.237	€ 400.000	€ 105.154	€ 265.000	€ 0	€ 2.000.000	€ 1.537.405	€ 3.537.405
Toekomstige projecten																
4	Nieuwe projecten															
5	aangevraagd binnen															
5	flagship															
...	Onderneming 2						€ 200.000		€ 400.000		€ 400.000		€ 265.000		€ 1.265.000	€ 1.265.000
...	Onderneming 3														€ 0	€ 0
...	Onderneming 4														€ 0	€ 0
...	Totale bijdrage ont	€ 0	€ 0	€ 0	€ 0	€ 0	€ 200.000	€ 0	€ 400.000	€ 0	€ 400.000	€ 0	€ 265.000	€ 0	€ 1.265.000	€ 1.265.000
Overige (private) partners																
7	Naam 1														€ 0	€ 0
8	Naam 2														€ 0	€ 0
9	Naam 3														€ 0	€ 0
...	Naam 4														€ 0	€ 0
...	Totale overige bidr	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0
Totaal budget per		€ 135.000	€ 159.669	€ 400.000	€ 543.638	€ 400.000	€ 646.706	€ 400.000	€ 882.237	€ 400.000	€ 505.154	€ 265.000	€ 265.000		Totaal	€ 4.802.405
Totaal budget		€ 294.669		€ 943.638		€ 1.046.706		€ 1.082.237		€ 905.154		€ 530.000		€ 4.272.405		
Funding gap															0-	

Annex 1: Detailed information matching funds

In kind contribution of iCELL applicants				
	Grant name	personnel	Total matching budget	matching duration
Chien	EMC-MOE (Taiwan)	PhD	96	4 years
Chien	Onco Institute	Postdoc (0.2 FTE), Technician (0,2FTE)	45	2 years
Kenjeres	Dutch Heart Foundation (Hartstichting) 'RADAR: Earlier Recognition of Aortic Atherosclerosis'	PhD (0.2 FTE)	35	2years
Kenjeres	FF4EuroHPC / DITAIDD - "A digital twin for airflow and inhaled drug delivery"	Senior Researcher (0,2FTE)	25	1 years
Pothof	NIH +ZONMW memorabel	PhD (0,5FTE), Postdoc (1FTE), technician	237	2 years
Koenderink	NWO-VICI How cytoskeletal teamwork makes cells strong	PhD (1FTE) +Consumables	145	2 years
Gribnau	NWO Psider	PhD + consumables	300	5 years
Brinks	ERC starting	3PhD (0,5; 0,75;0,75), consumables,	599	3 years
Goncalves		2PhD	250	2 years
van Meurs	ZonMW	0,5FTE +consumables	150	3 years
Adams		PhD (1FTE)	150	3 years
Redekop	AICCELERATE, Horizon 2020, Call: H2020-DT-2018-2020	Postdoc (0,3FTE)	105	2 years
Stankova	EU ETN project EvoGamesPlus	PhD (0,2FTE)	15	1 year
		TOTAL (Keuro)	2152	

II Letters of intent and letters of commitment



Addressed to:

Prof. Joyce van Meurs
Dr. Molewaterplein 40,
3015 GD Rotterdam, The Netherlands.
Email: j.vanmeurs@erasmusmc.nl

10th March 2022

Letter of Support for the project

Convergence in a dish: Individualized CELLular models of disease diversity in the population

Dear Professor van Meurs,

With this letter I declare on behalf of Mimetass our interest and support for the flagship project "Convergence in a dish: Individualized CELLular models of disease diversity in the population (iCELL)", as applied for by the main applicant, Prof. van Meurs, Principle Investigator at the departments of Internal Medicine and Orthopedics & Sportsmedicine.

As Mimetass, we have been fully informed about the above-mentioned application in the Flagship call within the Convergence initiative of the Erasmus MC, TU Delft and Erasmus University and we would like to express our interest in participating in this project and possible spin-off projects initiated within this flagship.

Mimetass is a private company, with more than 100 employees, with headquarters in Oegstgeest, the Netherlands, and subsidiaries in USA and Japan. MIMETAS strives to contribute to groundbreaking therapies with unique human disease biology, revealed by robust, screenable assays in versatile technology platforms. We develop novel phenotypic tissue models, tailored to support target discovery, lead optimization, and investigative tox.

The key to unlocking effective therapies for diseases is a deep understanding of human disease biology. To this end, MIMETAS aims to capture the essence of diseases in fit-for-purpose *in vitro* models. The goal of iCELL is to capture population diversity in diseases and response-to-treatment across the population in cell models using innovative technical solutions, and this goal is fully in line with the companies vision. Therefore, Mimetass participation in the iCELL flagship and possible spin-off projects is important for further development of the portfolio of Mimetass.

Yours sincerely,

Name: Jos Jodre, PhD

Position: CEO

Date: 10th March 2022

*This letter constitutes a statement of good faith intentions and a recital of support to the discussions in connection with the Convergence initiative Flagship call Proposal – "Convergence in a dish: Individualized CELLular models of disease diversity in the population", and therefore, is not intended to be legally binding upon the parties.

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Prof. Joyce van Meurs
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Date
11 March 2022

Our reference
SE.mb.22.001 / TNO_10026354

Page
1/2

Subject: Letter of support for the project
"Convergence in a dish: Individualized CELLular models of disease diversity in the population"

Dear Professor van Meurs,

With this letter I confirm on behalf of TNO our interest in and support for the flagship project "Convergence in a dish: Individualized CELLular models of disease diversity in the population (iCELL)", as applied for by the main applicant, Prof. van Meurs, Principle Investigator at the departments of Internal Medicine and Orthopedics & Sportsmedicine.

As TNO, the Netherlands Organisation for applied scientific research, we have been informed about the content of the above-mentioned application in the Flagship call within the Convergence initiative of the Erasmus MC, TU Delft and Erasmus University and we would like to express our interest in participating in this project and possible spin-off projects initiated within this flagship.

TNO, was founded to enable business and government to apply knowledge. As an organisation regulated by public law, we are independent: not part of any government, university or company.

TNO connects people and knowledge to create innovations that boost the competitive strength of industry and the well-being of society in a sustainable way.

The department of metabolic health research of TNO develops and performs in vitro and in vivo models to study mechanisms, interventions and biomarkers in metabolic diseases and their complications. These translational models include unique in vitro fibrosis models, read-out systems employing i.a. histology, biochemical assays, cell biology, molecular biology, immunology and inflammation markers. This preclinical research is strongly translational and aims to improve the predictability of efficacy and safety of pharmaceutical and food interventions by detailed knowledge of disease processes and mechanisms.



We look forward to collaborating with the flagship in the future.

Yours sincerely,

A blue digital signature of Sandra Eikhout, consisting of a stylized, cursive-like line.

Digitally signed
by Sandra
Eikhout, MSc.

SM Eikhout, MSc.
Director Market Healthy Living

Date
11 March 2022

Our reference
SE.mb.22.001 / TNO_10026354

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III Letter of support from department head of the main lead



Addressed to:
Prof. Joyce van Meurs
Dr. Molewaterplein 40,
3015 GD Rotterdam, The Netherlands.
Email: j.vanmeurs@erasmusmc.nl

8/3/2022

Letter of support for the project

Convergence in a dish: Individualized CELLular models of disease diversity in the population

Dear Prof. van Meurs,

With this letter I declare on behalf of the department of Internal Medicine our interest in and support for the flagship project "Convergence in a dish: Individualized CELLular models of disease diversity in the population (iCELL)", as applied for by the main applicant, Prof. van Meurs, Principal Investigator at the departments of Internal Medicine and Orthopedics & Sportsmedicine.

As head of the flagship leading department, I have been fully informed about the above-mentioned application in the Flagship call within the Convergence initiative of the Erasmus MC, TU Delft and Erasmus University. We are committed to participate in, and contribute to the proposed Flagship initiative.

The mission of the department of Internal Medicine is to pioneer innovation in research, education and care by bringing science to the patient. To achieve this objective, we aim to continually develop new technologies and innovations within the department in research, education and care. The goal of iCELL is to capture diversity in diseases and response-to-treatment across the population in cell models using innovative technical solutions. This goal is fully in line with the mission and strategy of the department of Internal Medicine: We have recently invested money to stimulate iPSC modelling of disease within the department showing relevant expertise and interest across our group. The department of Internal Medicine brings relevant contributions and expertise to this project and we therefore fully support the iCELL flagship.

Yours sincerely,

A handwritten signature in blue ink, appearing to be "R. Peeters", enclosed in a light blue rectangular box.

Name: Prof. dr Robin P. Peeters
Position: Head Department of. Internal Medicine
Date: 9/3/2022

Annex IV: References

1. Maciej S. Daniszewski, Anne Senabouth, Helena H. Liang et. al. Retinal ganglion cell-specific genetic regulation in primary open angle glaucoma. *BioRxiv* 2021; doi:<https://doi.org/10.1101/2021.07.14.452417>
2. Drew R. Neavin , Angela M. Steinmann, Han Sheng Chiu, et al. Village in a dish: a model system for population-scale hiPSC studies. *BioRxiv* 2021; doi: <https://doi.org/10.1101/2021.08.19.457030>
3. Li You , Pin-Rui Su , Max Betjes, et al. Functional single cell selection and annotated profiling of dynamically changing cancer cells. In press *Nature Biotechnology*. doi: <https://doi.org/10.1101/2021.10.12.464054>
4. Pin-Rui Su, Li You, Cecile Beerens, et al. Functional single cell proteomic profiling of cells with abnormal DNA damage response dynamics. *BioRxiv* 2021; doi: <https://doi.org/10.1101/2021.10.13.464241>
5. Martinez-Rivas A, González-Quijano GK, Proa-Coronado S, et al. Methods of Micropatterning and Manipulation of Cells for Biomedical Applications. *Micromachines* (Basel). 2017 Nov 29;8(12):347.
6. Doyle AD, Nazari SS, Yamada KM. Cell-extracellular matrix dynamics. *Phys Biol*. 2022 Jan 12;19(2):10.1088/1478-3975/ac4390.
7. Théry M. Micropatterning as a tool to decipher cell morphogenesis and functions. *J Cell Sci*. 2010 Dec 15;123(Pt 24):4201-13.
8. Vignaud T, Ennomani H, Théry M. Polyacrylamide hydrogel micropatterning. *Methods Cell Biol*. 2014;120:93-116.
9. Zafar, H., Wang, Y., Nakhleh, L. et al. Monovar: single-nucleotide variant detection in single cells. *Nat Methods* 13, 505–507 (2016).
10. Hao Y, Hao S, Andersen-Nissen E, et al. Integrated analysis of multimodal single-cell data. *Cell*. 2021 Jun 24;184(13):3573-3587.e29.
11. Ariel DH Gewirtz, F William Townes, Barbara E Engelhardt. Telescoping bimodal latent Dirichlet allocation to identify expression QTLs across tissues. *BioRxiv* 2021; doi: <https://doi.org/10.1101/2021.10.27.466156>
12. Squair, J.W., Gautier, M., Kathe, C. et al. Confronting false discoveries in single-cell differential expression. *Nat Commun* 12, 5692 (2021).
13. Julie Jerber, Daniel D. Seaton, Anna S.E. Cuomo et al. Population-scale single-cell RNA-seq profiling across dopaminergic neuron differentiation. *Nat Genet*. 2021 Mar 1; 53(3): 304–312.
14. <https://hpscereg.eu/collaborators/hipsci>
15. You, Li, Su, P.R., Betjes, M., et al. Linking the genotypes and phenotypes of cancer cells in heterogenous populations via real-time optical tagging and image analysis. *Nature Biomedical Engineering*, 2022. doi: 10.1038/s41551-022-00853-x.
16. Kateřina Staňková. Resistance games. *Nature Ecology & Evolution*. 3:336–337 (2019).
17. Leland McInnes, John Healy, James Melville. UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction. *Arxiv*. <https://doi.org/10.48550/arXiv.1802.03426>.
18. A. Moure and H. Gomez, Phase-field modeling of individual and collective cell migration. *Archives of Computational Methods in Engineering*, 28:311-344, 2021.
19. Elizabeth B.-N. Sanders and Pieter Jan Stappers. Co-creation and the new landscapes of design. *International Journal of CoCreation in Design and the Arts*. 4:5-18, 2008.

20. Jane Clemensen, Mette J Rothmann, Anthony C Smith et al. Participatory design methods in telemedicine research. *Journal of Telemedicine and Telecare*. 2017;23(9):780-785.
21. Katherine S Ruth, Felix R Day, Jazib Hussain et al., Genetic insights into biological mechanisms governing human ovarian ageing. *Nature*. 2021, Aug;596(7872):393-397.
22. Pin-Rui Su, Li You, Cecile Beerens, Karel Bezstarosti, et al., Functional single cell proteomic profiling of cells with abnormal DNA damage response dynamics. *BioRxiv* 2021 doi: <https://doi.org/10.1101/2021.10.13.464241>.